

SENDOV'S CONJECTURE HOLDS FOR EVERY DEGREE $n \geq 10^{200000}$

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ABSTRACT. Sendov's conjecture, first formulated in 1958, asserts that if a complex polynomial f of degree $n \geq 2$ has all its zeros in the closed unit disk $\{z \in \mathbb{C} : |z| \leq 1\}$, then, for every zero λ_0 of f , there exists a critical point ζ of f such that $|\zeta - \lambda_0| \leq 1$. The conjecture was previously known to hold for polynomials of degree $n \leq 8$, as well as in several special higher-degree cases. In 2022, using compactness methods, balayage, and the argument principle, Tao [Tao22] proved that there exists an absolute constant n_0 such that Sendov's conjecture holds for all $n \geq n_0$. However, Tao's argument does not provide an explicit admissible value of n_0 . In the present paper, we make Tao's result effective and prove that one may take

$$n_0 = 10^{200000}.$$

1. INTRODUCTION

For $z_0 \in \mathbb{C}$ and $r > 0$, write

$$D(z_0, r) = \{z \in \mathbb{C} : |z - z_0| < r\}, \quad \overline{D}(z_0, r) = \{z \in \mathbb{C} : |z - z_0| \leq r\},$$

and let $\mathbb{D} = D(0, 1)$.

In 1958, the Bulgarian mathematician Blagovest Sendov formulated the conjecture now known as *Sendov's conjecture*. It was subsequently included in Hayman's influential 1967 monograph *Research Problems in Function Theory* [Hay67, Problem 4.5].

Conjecture 1.1 (Sendov). *Let f be a complex polynomial of degree $n \geq 2$, all of whose zeros lie in $\overline{\mathbb{D}}$. If λ_0 is a zero of f , then f has a critical point in $\overline{D}(\lambda_0, 1)$.*

We first give a historical overview of the progress toward Conjecture 1.1.

- In 1968, Rubinstein [Rub68] proved Conjecture 1.1 when the distinguished zero λ_0 lies on the unit circle. In the same paper, he also verified the cases $n = 3, 4$.
- In 1969, Goodman, Rahman, and Ratti [GRR69] established a stronger localization result for a distinguished zero λ_0 on the unit circle, proving that f has a critical point ζ satisfying $|\zeta - \lambda_0/2| \leq 1/2$. In the same year, Joyal [Joy69] independently verified Conjecture 1.1 for distinguished zeros on the unit circle; namely, if $|\lambda_0| = 1$, then there exists a critical point ζ of f such that $|\zeta - \lambda_0| \leq 1$. Also in 1969, Meir and Sharma [MS69] proved Conjecture 1.1 for all polynomials of degree $n \leq 5$.
- In 1985, Bojanov, Rahman, and Szynal [BRS85] proved that if $f(z) = \prod_{j=1}^n (z - \lambda_j)$ has all its zeros in the closed unit disk, then, for every $\nu = 1, \dots, n$, the closed disk $\overline{D}\left(\lambda_\nu, \left(1 + \left|\prod_{j=1}^n \lambda_j\right|\right)^{1/n}\right)$ contains at least one zero of f' . Consequently, for each zero λ_ν of f , there exists a critical point ζ of f such that $|\zeta - \lambda_\nu| \leq 2^{1/n}$. Thus, they obtained Sendov's conclusion with the radius 1 replaced by the asymptotically sharp radius $2^{1/n} = 1 + \frac{\log 2}{n} + O(n^{-2})$.

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- In 1991, Brown [Bro91] proved the case $n = 6$.
- In 1993, Miller [Mil93] proved that for every fixed degree n , there exists a constant $\varepsilon_n > 0$, depending only on n , such that $|\lambda_0| \geq 1 - \varepsilon_n \implies$ there exists a critical point ζ of f satisfying $|\zeta - \lambda_0| \leq 1$. Thus, for each fixed degree n , Conjecture 1.1 holds throughout a sufficiently thin annulus adjacent to the unit circle. The argument was qualitative in the sense that it did not provide a usable explicit value of ε_n . Independently of Miller, Vâjăitu and Zaharescu [VZ93] proved that, for every fixed n , there exists $\varepsilon_n > 0$ such that $1 - \varepsilon_n \leq |\lambda_0| \leq 1 \implies$ there exists a critical point ζ of f satisfying $|\zeta - \lambda_0| \leq 1$. Their proof likewise did not yield an explicit numerical estimate for ε_n .
- In 1996, Borcea [Bor96] proved the case $n \leq 7$.
- In 1999, Brown and Xiang [BX99] proved the case $n \leq 8$.
- In 2010, Chijiwa [Chi10] initiated a quantitative version of the near-boundary analysis. Under the hypothetical assumption that a zero λ_0 is close to the unit circle but has no critical point within unit distance, he derived explicit estimates forcing the critical points to lie close to the origin and forcing the zeros of the polynomial to be close to a rotated configuration of the n -th roots of unity. These estimates provided the quantitative foundation for his explicit near-boundary theorem in the following year.
- In 2011, Chijiwa [Chi11] made the near-boundary result fully explicit. Let $n \geq 4$ and define $\varepsilon_n^C := \frac{1}{2n^{9/4}}$. If $|\lambda_0| \geq 1 - \varepsilon_n^C$, then there exists a critical point ζ such that

$$|\zeta - \lambda_0| \leq 1 - c_n(1 - |\lambda_0|),$$

where

$$c_n = \begin{cases} \frac{1}{4}, & n \equiv 0 \pmod{4}, \\ \frac{n-3}{4(n-1)}, & n \equiv 1 \pmod{4}, \\ \frac{n-6}{4(n-1)}, & n \equiv 2 \pmod{4}, \\ \frac{n-9}{4(n-1)}, & n \equiv 3 \pmod{4}. \end{cases}$$

In particular, together with the already known low-degree cases, this gives Conjecture 1.1 whenever the distinguished zero is exponentially close to the unit circle. In the same year, Bojanov [Boj11] showed that Conjecture 1.1 holds whenever $|\lambda_0| \leq \frac{1}{n-1}$. We note that the same conclusion follows immediately from the classical de Bruijn–Springer mean-modulus inequality [dBS47].

- In 2014, Kasmalkar [Kas14] enlarged Chijiwa’s admissible near-boundary region. For $n \geq 8$, define $\varepsilon_n^K := \frac{90}{n^{12} \log n}$. If $|\lambda_0| \geq 1 - \varepsilon_n^K$, then there exists a critical point ζ satisfying

$$|\zeta - \lambda_0| \leq 1 - c_n(1 - |\lambda_0|),$$

where c_n is the same explicit constant as in Chijiwa’s theorem. Thus Kasmalkar improved the required closeness from the exponentially small scale $\frac{1}{2n^{9/4}}$ to the polynomial scale $\frac{90}{n^{12} \log n}$. In the same year, Dégot [Deg14] proved a high-degree result for each fixed intermediate radius. More precisely, for every fixed $r \in (0, 1)$, there exists an integer $N(r)$ such that, whenever λ_0 is a zero of f with $|\lambda_0| = r$ and $n \geq N(r)$, there exists a critical

point ζ of f satisfying $|\zeta - \lambda_0| \leq 1$. The threshold in this theorem depends on the modulus of the distinguished zero and is therefore not uniform for $r \in (0, 1)$.

- In 2020, Chalebgwa [Cha20] made the dependence in Dégot's theorem explicit. If $0 < r = |\lambda_0| < 1$ and $n \geq \frac{20800}{r^7(1-r)^4}$, then there exists a critical point ζ of f satisfying $|\zeta - \lambda_0| \leq 1$. Equivalently, for a fixed degree n , this result covers the intermediate range in which r is neither too close to 0 nor too close to 1.
- In 2022, Tao [Tao22] proved that there exists an absolute constant n_0 such that Conjecture 1.1 holds for all $n \geq n_0$. His proof separates the argument into three regimes:
 - (1) when $|\lambda_0|$ is bounded away from both 0 and 1, it uses the results of Dégot and Chalebgwa;
 - (2) when $|\lambda_0|$ is close to 1, it refines Miller's near-boundary argument and invokes Chijiwa's quantitative theorem in the extremely close regime;
 - (3) when $|\lambda_0|$ is close to 0, it introduces a new argument based on compactness, balayage, logarithmic potentials, and the argument principle.

Because the proof uses a compactness-and-contradiction argument, it does not yield an explicit numerical value of n_0 .

In this paper, we make Tao's theorem for sufficiently large degrees [Tao22] effective. In Tao's theorem, one may take

$$n_0 = 10^{200000}.$$

Theorem 1.2. *Sendov's conjecture holds for every degree*

$$n \geq N_* := 10^{200000}.$$

For each fixed degree n , Sendov's conjecture can be formulated as a first-order sentence in the language of ordered fields. Hence, by Tarski's quantifier-elimination theorem [Tar31] for real closed fields, it is decidable in finite time. An explicit real-algebraic formulation, together with accompanying computational implementations, was given by Spjut [Spj20, Chapters 3–4].

Remark 1.3. Theorem 1.2 implies that Sendov's conjecture is effectively decidable. Indeed, it remains only to verify the finitely many degrees $2 \leq n < N_*$, and, for each such degree, the validity of the conjecture can be decided in finite time by quantifier elimination. Thus, applying the decision procedure successively to these finitely many degrees yields, in principle, a finite algorithm for deciding Sendov's conjecture in full.

Remark 1.4. The numerical bound in Theorem 1.2 can clearly be improved by refining the constants and parameter choices throughout the proof. Such an optimization is not essential to the present work; the main point is to obtain a completely explicit universal degree threshold.

Sketch of the proofs and new ideas. The magnitude of N_* in Theorem 1.2 reflects the deliberate use of broad numerical margins. The logical structure is nonetheless short. After rotation and multiplication by a nonzero scalar, a hypothetical counterexample has a monic polynomial f , a distinguished zero $a \in [0, 1]$, and no critical point in $\overline{D(a, 1)}$. Four ranges cover all $a > 0$:

$$\begin{array}{ll} 0 < a \leq n^{-1/8} & : \text{the effective near-origin argument of Section 3,} \\ a > n^{-1/8}, \quad 1 - a \geq n^{-1/64} & : \text{Chalebgwa's explicit theorem,} \\ a > n^{-1/8}, \quad 8^{-n} \leq 1 - a < n^{-1/64} & : \text{the effective boundary argument of Section 4,} \\ a > n^{-1/8}, \quad 0 \leq 1 - a < 8^{-n} & : \text{Chijiwa's explicit theorem.} \end{array}$$

The case $a = 0$ follows immediately from the Gauss–Lucas theorem [RS02, pp. 71–72, Theorem 2.1.1]. The proof of Theorem 1.2 is assembled in Section 5.

We now describe the two new quantitative ingredients. Near the origin, under the counterexample hypothesis in the regime $a \rightarrow 0$, Tao [Tao22, Theorem 1.10(i)] proved that the subsequential limiting laws of the zeros and critical points are both supported on the left semicircle C , and that these two limiting laws coincide. Instead of passing to a limiting measure, we compare the balayages of the zero and critical-point measures at two exterior radii. A Poisson-kernel test then places all but an explicitly bounded proportion of the zeros near the left semicircle. A fixed winding-number dichotomy and a selected circle complete the argument.

Near the unit circle, Tao's compactness argument [Tao22, Proposition 5.3] produces an at most countable exceptional set. We replace it by fewer than 10^{32} explicit obstacles. Away from a set of angles of total length less than 10^{-20} , an actual zero close to the prescribed boundary direction can be joined to the small disk by a path that stays a definite distance from every critical point. Integration of f''/f' along this path gives an effective level-set relation. The zeroth and second Fourier coefficients of the boundary potential then contradict the geometry of the arc $\{z : |z| \leq 1, |1 - z| = 1\}$.

Organization of the paper. Section 2 collects the basic identities and support properties for the zero and critical-point measures, and establishes an explicit comparison between their moments. Section 3 develops an effective argument for distinguished zeros near the origin, based on quantitative balayage estimates, harmonic-measure separation, and a winding-number argument. Section 4 treats distinguished zeros near the unit circle. Its main ingredients are a finite obstacle-avoidance construction, an effective level-set relation for the critical-point potential, and incompatible Fourier and geometric estimates for the second moment. In Section 5, the two endpoint arguments are combined with the quantitative results of Chalebgwa and Chijiwa to prove Theorem 1.2. Finally, Appendix A records the principal numerical inequalities and margins used throughout the proof.

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2. PRELIMINARIES

Let $n \geq 2$, and let f be monic of degree n , all of whose zeros lie in $\overline{\mathbb{D}}$. Let $\lambda_1, \dots, \lambda_n$ and $\xi_1, \dots, \xi_{n-1}$ denote the zeros of f and f' , respectively, counted with multiplicity. We write λ and ξ for uniformly chosen elements of these two multisets. Define

$$U_X(z) := \mathbb{E} \log \frac{1}{|z - X|}, \quad s_X(z) := \mathbb{E} \frac{1}{z - X}.$$

These functions are used only where the displayed expressions are finite.

Lemma 2.1 ([Tao22, Lemma 1.6, (ii)–(v)]). *For every monic polynomial f as above,*

$$\mathbb{E}\lambda = \mathbb{E}\xi.$$

Moreover,

$$U_\lambda(z) = -\frac{1}{n} \log |f(z)|, \quad (2.1)$$

$$U_\xi(z) = \frac{\log n - \log |f'(z)|}{n-1}, \quad (2.2)$$

$$s_\lambda(z) = \frac{f'(z)}{nf(z)}, \quad (2.3)$$

$$s_\xi(z) = \frac{f''(z)}{(n-1)f'(z)}. \quad (2.4)$$

Here (2.1) and (2.3) hold whenever $f(z) \neq 0$, whereas (2.2) and (2.4) hold whenever $f'(z) \neq 0$. If $f(z)f'(z) \neq 0$, then

$$U_\lambda(z) - \frac{n-1}{n} U_\xi(z) = \frac{1}{n} \log |s_\lambda(z)|. \quad (2.5)$$

The next elementary observation records the support restrictions imposed by a counterexample.

Lemma 2.2 ([Tao22, Lemma 1.6, (i), (ii)]). *Suppose that $f(a) = 0$, where $a \in [0, 1]$, and that f' has no zero in $\overline{D(a, 1)}$. Then*

$$|\xi| \leq 1, \quad |a - \xi| > 1$$

almost surely, and $\mathbb{E}\lambda = \mathbb{E}\xi$.

The following observation, due to Komarova and Rivin [KR03, Lemma 5.7], plays an important role in our derivation. It has also recently been used by Tang and the author in the study of Schoenberg-type inequalities [TZ25].

Lemma 2.3 ([KR03, Lemma 5.7]). *Let $n \geq 2$ and $\lambda_1, \lambda_2, \dots, \lambda_n \in \mathbb{C}$. Let $p(z) = \prod_{j=1}^n (z - \lambda_j)$, and define $D = \text{diag}(\lambda_1, \lambda_2, \dots, \lambda_n)$. Let I_n denote the $n \times n$ identity matrix, and let J_n denote the $n \times n$ matrix whose entries are all equal to 1. Then the characteristic polynomial of the matrix $D \left(I_n - \frac{1}{n} J_n \right)$ has the same multiset of zeros as the polynomial $zp'(z)$.*

Remark 2.4. Clearly, $I_n - \frac{1}{n} J_n$ is a projection matrix. Since AB and BA have the same spectrum for any $n \times n$ complex matrices A, B , all eigenvalues of $D \left(I_n - \frac{1}{n} J_n \right)$ coincide with those of $\left(I_n - \frac{1}{n} J_n \right) D \left(I_n - \frac{1}{n} J_n \right)$.

Tao's exterior-potential argument [Tao22, p. 363] gives the non-asymptotic moment comparison

$$|\mathbb{E}\lambda^m - \mathbb{E}\xi^m| = O\left(\frac{m \log m}{n}\right), \quad m \geq 2.$$

Lemma 2.5 below removes the logarithmic loss and makes the constant explicit, giving $5m/n$ for every $m \geq 1$. Its proof uses the D -companion matrix method rather than Tao's potential-theoretic argument.

Lemma 2.5. *For every integer $m \geq 1$,*

$$|\mathbb{E}\lambda^m - \mathbb{E}\xi^m| \leq \frac{5m}{n}. \quad (2.6)$$

Proof. Let $D = \text{diag}(\lambda_1, \dots, \lambda_n)$ and $P = I_n - \frac{1}{n} J_n$. By Lemma 2.3 and Remark 2.4, PDP has characteristic polynomial $zf'(z)/n$. Put $Q = I_n - P = \frac{1}{n} J_n$. Since Q has rank one and P, Q are orthogonal projections,

$$D - PDP = QD + PDQ.$$

Consequently,

$$\text{rank}(D - PDP) \leq \text{rank}(QD) + \text{rank}(PDQ) \leq 2.$$

Moreover, $\|D\| = \max_j |\lambda_j| \leq 1$, where $\|\cdot\|$ denotes the spectral norm. Therefore

$$\|D - PDP\| \leq \|QD\| + \|PDQ\| \leq 2\|D\| \leq 2.$$

The telescoping identity for matrix powers

$$D^m - (PDP)^m = \sum_{k=0}^{m-1} D^{m-1-k} (D - PDP) (PDP)^k$$

and $|\text{tr } A| \leq \text{rank}(A)\|A\|$ give

$$|\text{tr } D^m - \text{tr}(PDP)^m| \leq 4m.$$

Since

$$\text{tr}((PDP)^m) = \sum_{j=1}^{n-1} \xi_j^m \quad \text{and} \quad |\text{tr}((PDP)^m)| \leq n-1,$$

we obtain

$$\begin{aligned} |\mathbb{E}\lambda^m - \mathbb{E}\xi^m| &\leq \frac{1}{n} |\text{tr}(D^m) - \text{tr}((PDP)^m)| + \left| \frac{1}{n} - \frac{1}{n-1} \right| |\text{tr}((PDP)^m)| \\ &\leq \frac{4m+1}{n} \leq \frac{5m}{n}. \end{aligned}$$

This proves (2.6). $\square$

The preceding lemmas will be invoked in both endpoint regimes. We turn first to the origin, where a direct winding-number argument is available.

3. AN EFFECTIVE ARGUMENT NEAR THE ORIGIN

Throughout this section, f is monic of degree n , all of its zeros lie in $\overline{\mathbb{D}}$, and $a \in (0, 1)$ is a zero of f . We argue by contradiction and assume that

$$f'(\zeta) \neq 0 \quad \text{whenever} \quad |\zeta - a| \leq 1. \quad (3.1)$$

The notation λ, ξ, U_X, s_X is as in Section 2. Thus

$$s_\lambda(z) = \frac{f'(z)}{nf(z)}, \quad (3.2)$$

and Lemma 2.2 places ξ in the lune

$$\mathcal{L}_a := \{w \in \overline{\mathbb{D}} : |w - a| > 1\}$$

almost surely.

Tao [Tao22, Theorem 3.1] proved that there is no sequence of counterexamples of degrees $n_k \rightarrow \infty$ whose distinguished zeros a_k satisfy

$$a_k = o\left(\frac{1}{\log n_k}\right).$$

That result is asymptotic: its proof passes to a limiting root measure and selects an annulus from the zero set of the limiting Stieltjes transform, and therefore does not produce a numerical degree threshold.

The main theorem of this section is an effective specialization of Tao's result. It replaces the asymptotic hypothesis by the finite condition $0 < a \leq n^{-1/8}$ and supplies an explicit threshold $n \geq N_{\text{org}}$. Since $n^{-1/8} = o\left(\frac{1}{\log n}\right)$, this does not enlarge Tao's asymptotic range; the new

content is the uniform finite-degree argument with completely explicit constants. In particular, the nonconstructive limiting annulus is replaced below by a quantitative concentration estimate and a fixed winding-number dichotomy.

Theorem 3.1. *Set*

$$N_{\text{org}} := (9764480000000000000000)^{16}. \quad (3.3)$$

If $n \geq N_{\text{org}}$ and $0 < a \leq n^{-1/8}$, then f' has a zero in $\overline{D(a, 1)}$.

We begin its proof by making the comparison of balayages quantitative. For $R > 1$, $w \in \overline{\mathbb{D}}$, and $\theta \in \mathbb{R}$, put

$$P_w^R(Re^{i\theta}) := \frac{R^2 - |w|^2}{|Re^{i\theta} - w|^2}.$$

The normalization is

$$\int_{-\pi}^{\pi} P_w^R(Re^{i\theta}) \frac{d\theta}{2\pi} = 1.$$

For a random variable X taking values in $\overline{\mathbb{D}}$, define

$$\text{Bal}_R(X)(Re^{i\theta}) := \mathbb{E}P_X^R(Re^{i\theta}).$$

This is the density, with respect to normalized arclength measure on $\partial D(0, R)$, of the classical balayage of the law of X onto $\partial D(0, R)$; see [Gus04].

Tao [Tao22, p. 363, (2.9)] established the non-asymptotic balayage estimate

$$\|\text{Bal}_R(\lambda) - \text{Bal}_R(\xi)\|_{L^\infty(\partial D(0, R))} = O\left(\frac{\log(1/(R-1))}{n(R-1)^2}\right), \quad 1 < R \leq \frac{3}{2}.$$

The following lemma is a fully explicit version of this estimate. With $R = 1 + t$, it gives the constant 72 while preserving the same dependence on n and t . Its proof follows Tao's balayage–Fourier argument, with all implicit constants tracked quantitatively for later use.

Lemma 3.2. *Let $0 < t \leq 1/2$ and $R = 1 + t$. Then*

$$\|\text{Bal}_R(\lambda) - \text{Bal}_R(\xi)\|_{L^\infty(\partial D(0, R))} \leq \frac{72 \log(3/t)}{nt^2}.$$

Proof. Set $S = 1 + t/2$. If $z = Se^{i\theta}$, then

$$|s_\lambda(z)| \leq \frac{1}{S-1}$$

and

$$\Re(e^{i\theta} s_\lambda(Se^{i\theta})) = \mathbb{E} \Re \frac{1}{S - \lambda e^{-i\theta}} \geq \frac{1}{S+1}.$$

Consequently,

$$\frac{1}{S+1} \leq |s_\lambda(z)| \leq \frac{1}{S-1}.$$

Likewise,

$$-\log(S+1) \leq U_\xi(z) \leq -\log(S-1).$$

Using (2.5) and $S-1 = t/2$, we obtain

$$\|U_\lambda - U_\xi\|_{L^\infty(\partial D(0, S))} \leq \frac{2 \log(3/t)}{n}. \quad (3.4)$$

For $|w| \leq 1$, the absolutely convergent expansion

$$\log \frac{1}{|Se^{i\theta} - w|} = -\log S + \sum_{m=1}^{\infty} \frac{\Re(w^m e^{-im\theta})}{mS^m}$$

shows, on extracting the m -th Fourier coefficient in (3.4), that

$$|\mathbb{E}\lambda^m - \mathbb{E}\xi^m| \leq \frac{4mS^m \log(3/t)}{n} \quad (m \geq 1).$$

On the other hand,

$$\text{Bal}_R(X)(Re^{i\theta}) = 1 + 2\Re \sum_{m=1}^{\infty} \frac{\mathbb{E}X^m}{R^m} e^{-im\theta}.$$

It follows that

$$\|\text{Bal}_R(\lambda) - \text{Bal}_R(\xi)\|_{\infty} \leq \frac{8 \log(3/t)}{n} \sum_{m=1}^{\infty} m \left(\frac{S}{R}\right)^m.$$

Since $1 - \frac{S}{R} = \frac{t}{2(1+t)}$ and

$$\sum_{m=1}^{\infty} m \left(\frac{S}{R}\right)^m = \frac{S/R}{(1 - S/R)^2} \leq \frac{4(1+t)^2}{t^2} \leq \frac{9}{t^2},$$

the claimed bound follows. $\square$

We next quantify the differing harmonic-measure behavior of the left semicircle and of points separated from it. Put

$$C := \left\{ e^{i\phi} : \frac{\pi}{2} \leq \phi \leq \frac{3\pi}{2} \right\},$$

and, for $0 < \rho \leq 10^{-2}$, let

$$I_{\rho} := \left[-\frac{\pi}{2} + \frac{\rho}{4}, \frac{\pi}{2} - \frac{\rho}{4} \right].$$

We abbreviate

$$\omega_R(w, I_{\rho}) := \int_{I_{\rho}} P_w^R(Re^{i\theta}) \frac{d\theta}{2\pi}.$$

In the proof of [Tao22, Theorem 1.10(i)], Tao used a harmonic-measure separation between points bounded away from C and points in the thin lune $\mathcal{L}_a$. More precisely, for a fixed compact set $K \subset \overline{\mathbb{D}} \setminus C$ and a suitable fixed arc I , the harmonic measure of I is uniformly bounded below for starting points in K , whereas for starting points in $\mathcal{L}_a$ it is $O(a + R - 1)$. The following lemma makes the dependence on the separation scale ρ explicit. With $R = 1 + t$ and the arc I_{ρ} , it gives the respective bounds

$$\frac{\rho}{64} \quad \text{and} \quad \frac{250(a+t)}{\rho^2}.$$

Thus the underlying harmonic-measure argument is the same as Tao's, while all parameter dependence and numerical constants are made explicit.

Lemma 3.3. *Let $0 < \rho \leq 10^{-2}$, and suppose that*

$$0 < t \leq \frac{\rho}{4}, \quad 0 < a \leq \frac{\rho}{10}, \quad R = 1 + t.$$

Then the following estimates hold.

(i) *If $w \in \overline{\mathbb{D}}$ and $\text{dist}(w, C) \geq \rho$, then*

$$\omega_R(w, I_{\rho}) \geq \frac{\rho}{64}. \tag{3.5}$$

(ii) *If $w \in \mathcal{L}_a$, then*

$$\omega_R(w, I_{\rho}) \leq \frac{250(a+t)}{\rho^2}. \tag{3.6}$$

Proof. Write $w = re^{i\phi}$. If $r \leq 1 - \rho/2$, then

$$R^2 - r^2 \geq 1 - (1 - \rho/2)^2 \geq \frac{3\rho}{4}, \quad |Re^{i\theta} - w|^2 < (R + r)^2 < 5.$$

As $|I_\rho|/(2\pi) > 0.49$, this already proves (3.5).

Assume now that $r > 1 - \rho/2$. The separation from C forces $\phi \in (-\pi/2, \pi/2)$: otherwise $e^{i\phi} \in C$ and $|w - e^{i\phi}| = 1 - r < \rho$. Moreover,

$$\delta_\phi := \min \left\{ \phi + \frac{\pi}{2}, \frac{\pi}{2} - \phi \right\} > \frac{\rho}{2};$$

otherwise the nearer endpoint of C would be at distance at most $(1 - r) + \delta_\phi < \rho$ from w . Hence

$$[\phi - \rho/4, \phi + \rho/4] \subset I_\rho.$$

Put $s = R - r$ and $\eta = \min \left\{ \frac{s}{R}, \frac{\rho}{4} \right\}$. If $|\theta - \phi| \leq \eta$, then, using $1 - \cos u \leq u^2/2$, $r \leq R$, and $|\theta - \phi| \leq s/R$, we have

$$|Re^{i\theta} - re^{i\phi}|^2 = s^2 + 2Rr(1 - \cos(\theta - \phi)) \leq 2s^2$$

and $R^2 - r^2 = s(R + r) \geq sR$. Thus the Poisson kernel is at least $R/(2s)$. When $\eta = s/R$, integration gives $1/(2\pi)$. When $\eta = \rho/4$, the inequality

$$s < R - (1 - \rho/2) \leq 3\rho/4$$

gives instead $1/(6\pi)$. Both lower bounds are larger than $\rho/64$.

For the upper bound, write $w = x + iy$. The inequalities $|w| \leq 1$ and $|w - a| > 1$ imply

$$x < \frac{a}{2}, \quad 1 - |w|^2 < a^2 - 2ax \leq a^2 + 2a \leq 3a.$$

Therefore $R^2 - |w|^2 \leq 3(a + t)$. For $\theta \in I_\rho$,

$$R \cos \theta - x \geq \sin(\rho/4) - \frac{a}{2} \geq \frac{\rho}{5} - \frac{\rho}{20} = \frac{3\rho}{20}.$$

It follows that

$$P_w^R(Re^{i\theta}) \leq \frac{3(a + t)}{9\rho^2/400} < \frac{250(a + t)}{\rho^2}.$$

Integration proves (3.6). $\square$

The quantitative part of [Tao22, Theorem 1.10(i)] states that, for every fixed compact set

$$K \subset \overline{\mathbb{D}} \setminus C,$$

one has

$$\mathbb{P}(\lambda \in K) \ll_K a + \frac{\log n}{n^{1/3}}.$$

Thus, for each fixed $\rho > 0$, Tao's estimate already applies to

$$K_\rho := \{w \in \overline{\mathbb{D}} : \text{dist}(w, C) \geq \rho\}.$$

However, the implied constant in Tao's estimate is not made explicit as a function of ρ , and therefore cannot be used directly in the present finite-degree argument. Using the same balayage and harmonic-measure mechanism, the preceding lemmas give the fully explicit estimate

$$\mathbb{P}(\lambda \in K_\rho) \leq 73000\rho^{-3} \left(a + n^{-1/3} \log(3n) \right).$$

The point of the proposition below is therefore the explicit ρ^{-3} -dependence and the resulting effective numerical bound needed for the subsequent winding argument.

Proposition 3.4. *Let $\rho = 10^{-3}$, $t = n^{-1/3}$ and $R = 1 + t$. Assume $t \leq \rho/4$ and $a \leq \rho/10$. If $q := \mathbb{P}(\text{dist}(\lambda, C) \geq \rho)$, then*

$$q \leq 73000\rho^{-3} \left(a + n^{-1/3} \log(3n) \right).$$

Proof. Lemma 3.3 gives

$$\frac{\rho q}{64} \leq \mathbb{E}\omega_R(\lambda, I_\rho).$$

Lemmas 2.2, 3.3, and 3.2 give

$$\mathbb{E}\omega_R(\lambda, I_\rho) \leq \frac{250(a+t)}{\rho^2} + \frac{72 \log(3/t)}{nt^2}.$$

Since $t = n^{-1/3}$, this implies

$$q \leq 16000\rho^{-3}(a+t) + 4608\rho^{-1}t \log(3/t).$$

Now use $\rho < 1$, $\log(3/t) \leq \log(3n)$, and $\log(3n) > 1$, and enlarge the coefficient to 73000. $\square$

For the remainder of this section, fix

$$\rho := 10^{-3}, \quad q := \mathbb{P}(\text{dist}(\lambda, C) \geq \rho).$$

We record the numerical consequence that will be used below. Put $x = n^{1/16}$. If $n \geq N_{\text{org}}$, then

$$x \geq 9764480000000000000000 > 10^{22}, \quad \log n \leq x, \quad \log(3n) \leq 2x.$$

For $a \leq n^{-1/8}$, Proposition 3.4 therefore gives

$$\begin{aligned} q \log n &\leq 7.3 \cdot 10^{13} \left(n^{-1/8} + n^{-1/3} \log(3n) \right) \log n \\ &\leq 7.3 \cdot 10^{13} \left(x^{-1} + 2x^{-10/3} \right) < \frac{1}{4400000}. \end{aligned} \tag{3.7}$$

In particular, $q < 1$.

The remaining analytic input is a uniform winding alternative for measures on C .

Lemma 3.5. *Let ν be a probability measure supported on C , let W have law ν , and put*

$$\alpha = \mathbb{E}\overline{W}, \quad s_\nu(z) = \mathbb{E} \frac{1}{z - W}.$$

Then

$$s_\nu(z) = -\alpha + z + \mathcal{R}_\nu(z), \tag{3.8}$$

where, for $|z| < 1$,

$$|\mathcal{R}_\nu(z)| \leq 2|\alpha||z| + \frac{|z|^2}{1-|z|}. \tag{3.9}$$

If $|\alpha| \geq 1/20$, then for every $r \in [1/200, 1/100]$,

$$\min_{|z|=r} |s_\nu(z)| > 0.0388, \quad \text{wind}(s_\nu(\partial D(0, r)), 0) = 0.$$

If $|\alpha| < 1/20$, then for every $r \in [1/5, 21/100]$,

$$\min_{|z|=r} |s_\nu(z)| > 0.08, \quad \text{wind}(s_\nu(\partial D(0, r)), 0) = 1.$$

Proof. Since $|W| = 1$,

$$\frac{1}{z - W} = - \sum_{k=0}^{\infty} z^k \overline{W}^{k+1}.$$

For $W = e^{i\phi} \in C$,

$$|1 + \overline{W}^2| = 2|\cos \phi| = -2 \cos \phi.$$

Thus

$$\mathbb{E}|1 + \overline{W}^2| = -2\Re \alpha \leq 2|\alpha|.$$

Adding and subtracting z in the power series proves (3.8) and (3.9).

If $|\alpha| \geq 1/20$ and $r \leq 1/100$, then

$$|s_\nu(z)| \geq |\alpha|(1 - 2r) - r - \frac{r^2}{1 - r} \geq \frac{1}{20} \cdot 0.98 - 0.01 - \frac{0.01^2}{0.99} > 0.0388.$$

The same estimate gives $|s_\nu(z) + \alpha| < |\alpha|$, so the straight-line homotopy to $-\alpha$ avoids zero and has winding number 0.

If $|\alpha| < 1/20$ and $1/5 \leq r \leq 21/100$, then

$$|s_\nu(z) - z| \leq \frac{1}{20}(1 + 2r) + \frac{r^2}{1 - r} < r.$$

Indeed, the function $r - \frac{1}{20}(1 + 2r) - \frac{r^2}{1 - r}$ has derivative

$$\frac{9}{10} - \frac{r(2 - r)}{(1 - r)^2} > 0$$

on $[1/5, 21/100]$, and it equals 0.08 at $r = 1/5$. Thus the margin is at least 0.08 on the stated interval. Rouché's theorem compares s_ν with z , giving winding number 1. $\square$

It remains to choose a boundary circle that avoids the zeros of f and on which the exceptional zeros contribute little. Let

$$G := \{\text{dist}(\lambda, C) < \rho\}.$$

By (3.7), $q < 1$, so $\mathbb{P}(G) = 1 - q > 0$. Choose a Borel nearest-point projection $\pi : G \rightarrow C$, so that $|\lambda - \pi(\lambda)| < \rho$ on G , and let ν be the conditional law of $\pi(\lambda)$ given G . The exact decomposition is

$$\begin{aligned} s_\lambda(z) &= (1 - q)s_\nu(z) + E_{\text{proj}}(z) + E_{\text{bad}}(z), \\ E_{\text{proj}}(z) &:= \mathbb{E} \left[\mathbb{1}_G \left(\frac{1}{z - \lambda} - \frac{1}{z - \pi(\lambda)} \right) \right], \\ E_{\text{bad}}(z) &:= \mathbb{E} \frac{\mathbb{1}_{G^c}}{z - \lambda}. \end{aligned}$$

The radius-selection mechanism used below already appears, in non-explicit form, in the proof of [Tao22, Theorem 3.1]. There Tao splits off the exceptional zeros, truncates the singular kernel at the scale n^{-10} , and averages over an interval of radii. The Fubini–Tonelli theorem and Markov's inequality then show that the exceptional contribution is small for most radii, after which small intervals around the moduli of the zeros of f are deleted to ensure that the selected circle contains no zero.

The following lemma applies the same selection principle to the decomposition above and makes every estimate explicit. The exceptional contribution E_{bad} is bounded by $8800q \log n$, while the additional error E_{proj} , arising from projection onto C , is bounded by 0.00161. The new idea is the explicit finite-degree form and the accompanying projection estimate, rather than the underlying Fubini–Markov radius-selection argument.

Lemma 3.6. *Assume $n \geq 3$. Let J be either*

$$J_0 = [1/200, 1/100] \quad \text{or} \quad J_1 = [1/5, 21/100].$$

There is $r \in J$ such that f has no zero on $\partial D(0, r)$ and

$$\sup_{|z|=r} |E_{\text{bad}}(z)| \leq 8800q \log n.$$

For this r ,

$$\sup_{|z|=r} |E_{\text{proj}}(z)| < 0.00161.$$

Proof. Put $\delta = n^{-10}$ and

$$X(r) := \mathbb{E} \frac{\mathbb{1}_{G^c}}{\max\{|r - |\lambda||, \delta\}}.$$

For every $s \in [0, 1]$ and for either interval J ,

$$\int_J \frac{dr}{\max\{|r - s|, \delta\}} \leq 2 + 2 \log(1/\delta) \leq 22 \log n.$$

Fubini's theorem gives

$$\int_J X(r) dr \leq 22q \log n.$$

Writing $\ell = |J| \geq 1/200$, the set on which

$$X(r) > \frac{44q \log n}{\ell}$$

has measure at most $\ell/2$. Delete also the intervals of radius δ centered at the moduli of the n zeros of f . Their total length is at most $2n\delta = 2n^{-9} < \ell/2$. A point r remains. At that point f has no boundary zero and

$$\sup_{|z|=r} |E_{\text{bad}}(z)| \leq X(r) \leq \frac{44q \log n}{\ell} \leq 8800q \log n.$$

Finally, $r \leq 0.21$, $\rho = 0.001$, and $|\lambda - \pi(\lambda)| < \rho$ on G , so

$$|E_{\text{proj}}(z)| \leq \frac{\rho}{(1-r-\rho)(1-r)} \leq \frac{0.001}{0.789 \cdot 0.79} < 0.00161.$$

□

All ingredients are now in place to prove the endpoint theorem.

Proof of Theorem 3.1. Assume (3.1). Set $\rho = 10^{-3}$ and $t = n^{-1/3}$. The hypotheses imply

$$t \leq \rho/4, \quad a \leq \rho/10,$$

and (3.7) gives

$$8800q \log n \leq \frac{1}{500}.$$

If $|\alpha| \geq 1/20$, use J_0 in Lemma 3.6; otherwise use J_1 . On the selected circle,

$$|s_\lambda(z) - (1-q)s_\nu(z)| < 0.00161 + 0.002 = 0.00361.$$

By Lemma 3.5,

$$|(1-q)s_\nu(z)| > \begin{cases} (1-q)0.0388, & |\alpha| \geq 1/20, \\ (1-q)0.08, & |\alpha| < 1/20. \end{cases}$$

Since $q < 1/4400000$, both lower bounds exceed 0.00361. The straight-line homotopy between s_λ and $(1-q)s_\nu$ consequently avoids zero, and

$$\text{wind}(s_\lambda(\partial D(0, r)), 0) \in \{0, 1\}. \quad (3.10)$$

The selected radius satisfies $r \leq 0.21$, whereas $a \leq n^{-1/8} < 10^{-3}$. Hence

$$\overline{D(0, r)} \subset D(a, 1),$$

because $r + a < 0.211 < 1$. By (3.1), f' has no zero in $\overline{D(0, r)}$, and therefore

$$N_{D(0, r)}(f') = 0.$$

The construction of r also ensures that f has no zero on $\partial D(0, r)$, while $a < 1/200 \leq r$ places at least one zero of f inside. Writing $N_\Omega(g)$ for the number of zeros of g in Ω , counted with multiplicity, the argument principle applied to (3.2) gives

$$\text{wind}(s_\lambda(\partial D(0, r)), 0) = N_{D(0, r)}(f') - N_{D(0, r)}(f) \leq -1,$$

contradicting (3.10). $\square$

The argument near the origin is thus fully effective. The boundary endpoint requires additional geometry because the limiting critical-point support may have finitely many macroscopic outliers.

4. AN EFFECTIVE ARGUMENT NEAR THE UNIT CIRCLE

Let $n \geq N_*$, and suppose again that f is monic, all its zeros lie in $\overline{\mathbb{D}}$, $f(a) = 0$, and

$$f'(\zeta) \neq 0 \quad (|\zeta - a| \leq 1). \quad (4.1)$$

After rotation, $a \in (0, 1)$. Put

$$d = 1 - a \quad \text{and assume} \quad 8^{-n} \leq d \leq n^{-1/64}. \quad (4.2)$$

For the random variables λ and ξ , write

$$\mu = \mathbb{E}\lambda = \mathbb{E}\xi, \quad \sigma^2 = \mathbb{E}|\xi - \mu|^2, \quad V = \mathbb{E}|\xi|^2 = \sigma^2 + |\mu|^2.$$

By Lemma 2.2,

$$|\xi| \leq 1, \quad |a - \xi| > 1 \quad (4.3)$$

almost surely.

The first lemma supplies the quantitative small parameter used throughout this section. The polynomial inequalities used in its proof are due to Dégot [Deg14, Lemma 1 and Theorem 4].

Lemma 4.1. *Under (4.1) and (4.2),*

$$|f(0)| > \frac{1}{16}, \quad n < |f'(a)| \leq 16n. \quad (4.4)$$

If $L = \log |a - \xi|$, then

$$0 < L \leq \log 2, \quad \mathbb{E}L < \frac{4}{n}, \quad (4.5)$$

and

$$V < 5n^{-1/64} < 10^{-3124}. \quad (4.6)$$

In particular,

$$|\mu|, \sigma < 10^{-1562}. \quad (4.7)$$

Proof. Since $f'(a) = n \prod_{j=1}^{n-1} (a - \xi_j)$, (4.3) gives $|f'(a)| > n$. For $0 < t < a$, Dégot's estimates [Deg14, Theorem 4, Lemma 1] give

$$|f(t)| \geq \frac{1 - \sqrt{1 + t^2 - at}}{n} |f'(a)|, \quad (4.8)$$

$$|f(t)| \leq (1 + t^2 - 2t\Re\mu)^{n/2}. \quad (4.9)$$

The range (4.2) implies $a > 0.99$. If $0 < t \leq 0.1$, then

$$1 - \sqrt{1 + t^2 - at} = \frac{t(a - t)}{1 + \sqrt{1 - t(a - t)}} > 0.44t.$$

Combining (4.8) and (4.9) with $|f'(a)| > n$, taking logarithms, and using $\log(1+x) \leq x$, yields

$$\Re \mu \leq B(t) := \frac{t}{2} + \frac{\log(1/(0.44t))}{nt}. \quad (4.10)$$

At $t = n^{-1/3}$, the right side is less than 0.01.

Write $\xi = u + iv$. From (4.3),

$$u < \frac{|\xi|^2 + a^2 - 1}{2a} \leq \frac{a}{2}.$$

Since $u \geq -1$ and $\mathbb{E}u < 0.01$, we have

$$\begin{aligned} \mathbb{E}u &\geq -\mathbb{P}(u \leq 0.2) + 0.2\mathbb{P}(u > 0.2) \\ &= -\mathbb{P}(u \leq 0.2) + 0.2(1 - \mathbb{P}(u \leq 0.2)) \\ &= 0.2 - 1.2\mathbb{P}(u \leq 0.2). \end{aligned}$$

It follows that

$$0.2 - 1.2\mathbb{P}(u \leq 0.2) < 0.01,$$

and therefore

$$\mathbb{P}(u \leq 0.2) > \frac{19}{120}. \quad (4.11)$$

For $0 \leq x \leq 1/2$, the support condition $u < a/2$ gives

$$|a - \xi|^2 - |x - \xi|^2 = (a - x)(a + x - 2u) \geq 0.$$

Thus

$$|x - \xi| \leq |a - \xi|$$

for every critical point ξ . On the event $u \leq 0.2$, set

$$F_a(x) := (a - x)(a + x - 0.4) = a^2 - 0.4a + 0.4x - x^2.$$

Since F_a is concave on $[0, 1/2]$, its minimum on this interval is attained at an endpoint. Using $a > 0.99$, we obtain

$$F_a(0) = a(a - 0.4) \geq 0.99(0.99 - 0.4) = 0.5841$$

and

$$F_a(1/2) = (a - 0.5)(a + 0.1) \geq (0.99 - 0.5)(0.99 + 0.1) = 0.5341.$$

Consequently,

$$F_a(x) \geq 0.5341 \quad (0 \leq x \leq 1/2).$$

Since $|x - \xi| \leq x + |\xi| \leq 3/2$, it follows that

$$\frac{|a - \xi|^2}{|x - \xi|^2} = 1 + \frac{(a - x)(a + x - 2u)}{|x - \xi|^2} \geq 1 + \frac{0.5341}{(3/2)^2}.$$

Therefore

$$\log \frac{|a - \xi|}{|x - \xi|} \geq \frac{1}{2} \log \left(1 + \frac{0.5341}{(3/2)^2} \right) > 0.1 \quad \text{on } \{u \leq 0.2\}.$$

Using the factorization of f' and (4.11), we obtain

$$|f'(x)| \leq e^{-(n-1)/100} |f'(a)| \quad (0 \leq x \leq 1/2). \quad (4.12)$$

At $x = 1/2$, (4.8) gives

$$|f(1/2)| > \frac{|f'(a)|}{8n}.$$

Integrating (4.12) on $[0, 1/2]$, and noting that $8ne^{-(n-1)/100} < 1$, we find

$$|f(0)| \geq |f(1/2)| - \frac{1}{2}e^{-(n-1)/100}|f'(a)| > \frac{|f'(a)|}{16n}.$$

Since $|f(0)| = \prod_j |\lambda_j| \leq 1$, this proves (4.4).

The factorization of f' now gives

$$(n-1)\mathbb{E}L = \log \frac{|f'(a)|}{n} < \log 16,$$

and hence (4.5). Moreover,

$$e^{2L} = |a - \xi|^2 = a^2 - 2a\Re\xi + |\xi|^2,$$

so

$$V = 2a\Re\mu + \mathbb{E}e^{2L} - a^2.$$

The chord joining the endpoints of the convex function e^{2x} on $[0, \log 2]$ gives

$$e^{2L} \leq 1 + \frac{3}{\log 2}L.$$

Consequently,

$$V \leq 2a\Re\mu + (1 - a^2) + \frac{18}{n}.$$

Apply (4.10) with $t = n^{-1/64}$. Here

$$\log \frac{1}{0.44t} < \log(3n) < n^{1/16} < \frac{1}{2}n^{31/32},$$

so $B(t) < t$. Since $d \leq t$, it follows that

$$V < 2t + 2d + \frac{18}{n} < 5n^{-1/64}.$$

Since $n \geq 10^{200000}$, the last expression is less than 10^{-3124} . Equation (4.7) follows from $|\mu|^2, \sigma^2 \leq V$. $\square$

The critical points are now close to a fixed circular arc, apart from a uniformly bounded number of exceptions. Define

$$\mathcal{A} = \{z \in \mathbb{C} : |z| \leq 1, |1 - z| = 1\}.$$

The following lemma should be compared with [Tao22, pp. 389–390].

Lemma 4.2. *For every zero ξ of f' , there is $y \in \mathcal{A}$ such that*

$$|\xi - y| \leq 6(d + L), \quad L = \log |a - \xi|.$$

Proof. Put $r = |a - \xi| = e^L$ and $R = |1 - \xi|$. Then

$$|R - 1| \leq |R - r| + (r - 1) \leq d + 2L,$$

because $e^L - 1 \leq 2L$ on $[0, \log 2]$. Radially project ξ , with center 1, to the circle $|1 - z| = 1$, obtaining y_0 . Then $|\xi - y_0| \leq d + 2L$.

If $|y_0| \leq 1$, take $y = y_0$. Otherwise write $y_0 = 1 + e^{i\phi}$ and take the nearer endpoint of $\mathcal{A}$. The elementary identities

$$|y_0| = 2|\cos(\phi/2)|, \quad |e^{i\phi} - e^{\pm 2\pi i/3}| = 2\sin\left|\frac{\phi \mp 2\pi/3}{2}\right|$$

show directly that

$$\text{dist}(y_0, \mathcal{A}) \leq 2(|y_0| - 1) = 2 \text{dist}(y_0, \overline{\mathbb{D}}) \leq 2|\xi - y_0|.$$

The triangle inequality now gives

$$\text{dist}(\xi, \mathcal{A}) \leq 3(d + 2L) \leq 6(d + L).$$

□

We fix, once and for all, the parameters

$$\begin{aligned} \Delta &= 10^{-20}, & g &= 10^{-22}, & \tau &= 10^{-30}, & K_0 &= 10^{32}, \\ h &= 10^{-110}, & r_* &= 2.5 \cdot 10^{-172}, & d_* &= 2 \cdot 10^{-172}, & \kappa &= 10^{-400}. \end{aligned} \quad (4.13)$$

Let T be the multiset of critical points t for which $\text{dist}(t, \mathcal{A}) > \tau$.

The path construction in the following lemma is a quantitative version of the obstacle-avoidance argument used by Tao in the proofs of [Tao22, Propositions 5.3 and 5.4]. In place of the compact exceptional set arising in Tao's asymptotic argument, we work with finitely many explicit obstacles and construct paths starting from actual zeros rather than ideal boundary points. The lemma also gives an explicit bound for the number of obstacles, collects all forbidden directions into a single exceptional set H , and ensures that the resulting admissible paths remain a fixed positive distance from every obstacle.

Lemma 4.3. *The multiset T has fewer than K_0 elements. There is a measurable set $H \subset \mathbb{R}/(2\pi\mathbb{Z})$, invariant under translation by π , with*

$$|H| < \Delta$$

and the following property. If $\theta \notin H$ and z is a zero of f satisfying

$$|z - e^{i\theta}| \leq 4\kappa, \quad (4.14)$$

then z can be joined to a point w_0 , $|w_0| = 0.01$, by a piecewise smooth path γ of length less than 1.3 such that, with s denoting arclength from z ,

$$-\frac{d}{ds}|\gamma(s)| \geq 0.8 \quad \text{for a.e. } s \in (0, \ell(\gamma)), \quad (4.15)$$

and

$$\text{dist}(\gamma, \text{supp } \xi) > d_*. \quad (4.16)$$

Proof. If $t \in T$, Lemma 4.2 gives $\tau < 6(d + L_t)$. Since $d < \tau/10$, this implies $L_t > \tau/20$. On the other hand,

$$\sum_{j=1}^{n-1} L_j = \log \frac{|f'(a)|}{n} < \log 16.$$

Consequently,

$$\#T < \frac{20 \log 16}{\tau} < K_0.$$

For $u \in \mathbb{R}$, put $F_u(w) = 2\Re w - u|w|^2$. Since $0 \in \mathcal{A}$, every $t \in T$ is nonzero and $|t| > \tau$; set $u_t = 2\Re t/|t|^2$. Delete every angle satisfying one of

$$|2 \cos \theta - 1| \leq 2g, \quad (4.17)$$

$$|2 \cos \theta - u_t| \leq 2h \quad (t \in T), \quad (4.18)$$

$$|\sin(\theta - \arg t)| \leq \frac{2r_*}{|t|} \quad (t \in T). \quad (4.19)$$

Enlarge this set by a 100κ -neighborhood and take its union with its translate by π . For $u, \alpha \in \mathbb{R}$ and $r \geq \tau$, the following one-dimensional estimates hold:

$$\begin{aligned} |\{2 \cos \theta - 1 \leq 2g\}| &< 20g, \\ |\{2 \cos \theta - u \leq 2h\}| &< 40\sqrt{h}, \\ |\{|\sin(\theta - \alpha)| \leq 2r_*/r\}| &< 32r_*/r. \end{aligned}$$

For completeness, the first set is contained in two intervals on which $|\sin \theta| > 0.8$. The mean-value theorem therefore gives

$$|\{\theta : |2 \cos \theta - 1| \leq 2g\}| < 20g.$$

For the second estimate, remove the $\sqrt{h}$ -neighborhoods of the two turning points 0 and π . Their total length is at most $4\sqrt{h}$. On the complement,

$$|(2 \cos \theta)'| = 2|\sin \theta| \geq \sqrt{h}.$$

Applying the mean-value theorem on the two monotonicity intervals of $\cos \theta$ gives

$$|\{\theta : |2 \cos \theta - u| \leq 2h\}| < 40\sqrt{h}.$$

Finally, if $0 < \eta < 1/4$, then

$$|\{\theta : |\sin(\theta - \alpha)| \leq \eta\}| = 4 \arcsin \eta \leq 8\eta.$$

Taking $\eta = 2r_*/r$, which lies in $(0, 1/4)$ because $r \geq \tau$, yields

$$|\{\theta : |\sin(\theta - \alpha)| \leq 2r_*/r\}| < 32r_*/r.$$

After the π -symmetrization and thickening, these estimates give the following explicit bound. Before symmetrization the deleted set has at most $4 + 8K_0$ interval components; enlarging each component by 100κ adds at most $200\kappa(4 + 8K_0) < 2000(K_0 + 1)\kappa$ to its length. Therefore

$$|H| < 2 \left(20g + 40K_0\sqrt{h} + \frac{32K_0r_*}{\tau} + 2000(K_0 + 1)\kappa \right) < 6 \cdot 10^{-21} < \Delta.$$

Let $\theta \notin H$ and let z satisfy (4.14). Put $u_z = 2\Re z/|z|^2$. The map $w \mapsto 2\Re w/|w|^2$ is 10-Lipschitz on $\{|w| \geq 0.99\}$; hence

$$|u_z - 2 \cos \theta| < 50\kappa. \tag{4.20}$$

The deletion (4.17) implies that precisely one of the following alternatives applies:

$$u_z \leq 0, \quad u_z \geq 1 + \frac{g}{3}, \quad 0 < u_z \leq 1 - \frac{g}{3}.$$

In the first two alternatives, take the radial segment from z to the circle $|w| = 0.01$. In the third alternative, $0 < u_z \leq 1 - g/3$ and $F_{u_z}(z) = 0$. In polar coordinates $w = re^{i\phi}$, the nonzero part of the level set $F_{u_z}(w) = 0$ is given by

$$r = \frac{2 \cos \phi}{u_z}, \quad |\phi| < \frac{\pi}{2}.$$

Starting at z , move toward $+\pi/2$ when $\arg z \geq 0$, and toward $-\pi/2$ when $\arg z < 0$, until $r = 0.01$. Along this branch r decreases monotonically, so the whole arc lies in $\mathbb{D}$. Since $F_{u_z}(w) = 0$ is the circle centered at $1/u_z$ with radius $1/u_z$, differentiation with respect to arclength gives

$$-\frac{dr}{ds} = \sqrt{1 - \frac{u_z^2 r^2}{4}} \geq \frac{\sqrt{3}}{2},$$

where we used $u_z r \leq 1$. Thus each chosen path has length less than

$$\frac{1 - 0.01}{\sqrt{3}/2} < 1.3$$

and satisfies (4.15).

On a circular path,

$$|F_1(w)| = |1 - u_z||w|^2 \geq \frac{g}{3} 10^{-4}.$$

If the radial path has $u_z \leq 0$, then

$$|F_1(re^{i \arg z})| = r(r - u_z|z|) \geq r^2.$$

If it has $u_z \geq 1 + g/3$, then

$$F_1(re^{i \arg z}) = r(u_z|z| - r) \geq \frac{g}{3} r|z|$$

for $0.01 \leq r \leq |z|$. Thus the same lower bound holds on every path. Since $F_1 = 0$ on $\mathcal{A}$ and $|\nabla F_1| \leq 4$ in $\mathbb{D}$,

$$\text{dist}(w, \mathcal{A}) > 5 \cdot 10^{-28} \quad (w \in \gamma). \quad (4.21)$$

For a circular path and $t \in T$, (4.18) and (4.20) give

$$|F_{u_z}(t)| = |u_z - u_t||t|^2 > h\tau^2 = 10^{-170}.$$

Here $|\nabla F_{u_z}| \leq 4$ on the unit disk, so

$$\text{dist}(t, \gamma) > 2.5 \cdot 10^{-171} > d_*.$$

For a radial path, the 100κ -thickening and (4.19), together with $|\arg z - \theta| < 5\kappa$, give

$$\begin{aligned} |\sin(\arg z - \arg t)| &\geq |\sin(\theta - \arg t)| - |\arg z - \theta| \\ &> \frac{2r_*}{|t|} - 5\kappa > \frac{r_*}{|t|}. \end{aligned}$$

Thus the distance from t to the full line containing the radial segment, and hence to the segment itself, is greater than $r_* > d_*$. Finally, if $t \in \text{supp } \xi \setminus T$ and $w \in \gamma$, then $\text{dist}(t, \mathcal{A}) \leq \tau$, and hence

$$|w - t| \geq \text{dist}(w, \mathcal{A}) - \text{dist}(t, \mathcal{A}) > 5 \cdot 10^{-28} - \tau > d_*$$

by (4.21). This proves (4.16). $\square$

We next show that every boundary direction has a nearby zero of f ; for directions outside H , Lemma 4.3 then applies to that zero. The discrepancy estimate used here is the Fourier form of the Erdős–Turán inequality [Mon94, Corollary 1.1].

Lemma 4.4. *For every $\theta \in \mathbb{R}$, there is a zero z of f such that*

$$|z - e^{i\theta}| < 4\kappa. \quad (4.22)$$

Proof. By (4.4),

$$\sum_{j=1}^n (1 - |\lambda_j|) \leq \sum_{j=1}^n \log \frac{1}{|\lambda_j|} = \log \frac{1}{|f(0)|} < \log 16 < 3. \quad (4.23)$$

In particular, no λ_j is zero. Put $\omega_j = \lambda_j/|\lambda_j|$, and let ν be their empirical angular measure. We use the convention

$$\hat{\nu}(m) := \int_{\partial \mathbb{D}} w^m d\nu(w) = \frac{1}{n} \sum_{j=1}^n \omega_j^m, \quad m \geq 1.$$

Lemma 2.5 and (4.23) give

$$|\hat{\nu}(m) - \mathbb{E}\xi^m| \leq \frac{8m}{n} \quad (m \geq 1).$$

Moreover,

$$|\mathbb{E}\xi| \leq \sqrt{V}, \quad |\mathbb{E}\xi^m| \leq V \quad (m \geq 2).$$

In the normalization used here, the Erdős–Turán inequality [Mon94, Corollary 1.1] is

$$D(\nu) \leq 10 \left\{ M^{-1} + \sum_{m=1}^M \frac{|\hat{\nu}(m)|}{m} \right\}.$$

With $M = 10^{410}$, it therefore gives

$$D(\nu) \leq 10 \left(M^{-1} + \sqrt{V} + V \log M + \frac{8M}{n} \right) < \frac{\kappa}{100\pi}.$$

Here $D(\nu)$ is the supremum, over arcs I , of $|\nu(I) - |I|/(2\pi)|$.

The arc centered at θ and having angular length 4κ contains more than $0.6n\kappa$ zero arguments. By (4.23), fewer than $3/\kappa$ zeros have radial defect larger than κ . Since $n > 5\kappa^{-2}$, the centered arc contains a zero with radial defect at most κ . Its angular displacement is at most 2κ , so its Euclidean distance from $e^{i\theta}$ is less than 3κ , proving (4.22). $\square$

The geometric preparation is complete. We next integrate the differential equation for f' along the separated paths.

Lemma 4.5. *Let q be either a or a zero z supplied by Lemma 4.4 for an angle outside H . Then*

$$0 = f(0) + \frac{f'(q)(q - \mu)}{n}(1 + \vartheta_q) + E_q, \quad (4.24)$$

where

$$|\vartheta_q| \leq 10^{182}\sigma^2, \quad |E_q| < 0.03^n. \quad (4.25)$$

Furthermore,

$$|\log |f(0)| + nU_\xi(q)| \leq 3 \cdot 10^{182}\sigma^2 + 32(0.06)^n. \quad (4.26)$$

Proof. For $q = z$, use the path in Lemma 4.3. For $q = a$, use the positive real segment from a to 0.01. Indeed, for $0.01 \leq x \leq a$,

$$|x - \xi| \geq |a - \xi| - |a - x| > x + d \geq 0.01.$$

Along either path, $|w| \geq 0.01$ and $\text{dist}(w, \text{supp } \xi) \geq d_*$.

Substituting $x = w - \mu$ and $y = \xi - \mu$ into the algebraic identity

$$\frac{1}{x - y} = \frac{1}{x} + \frac{y}{x^2} + \frac{y^2}{x^2(x - y)}$$

and then taking expectations gives

$$s_\xi(w) = \frac{1}{w - \mu} + \mathbb{E} \frac{(\xi - \mu)^2}{(w - \mu)^2(w - \xi)}.$$

Equations (4.7) and (4.16) imply

$$\left| s_\xi(w) - \frac{1}{w - \mu} \right| < 10^{178}\sigma^2.$$

If the path is parametrized by arclength from q , then (4.15) applies when $q = z$, whereas $-d|\gamma(s)|/ds = 1$ when $q = a$. Differentiating $|\gamma(s) - \mu|$, and using

$$\left| \frac{d}{ds} |\gamma(s) - \mu| - \frac{d}{ds} |\gamma(s)| \right| \leq \frac{2|\mu|}{|\gamma(s)| - |\mu|} < 0.01,$$

therefore gives

$$-\frac{d}{ds}|\gamma(s) - \mu| > 0.79.$$

As $0.989 < |q - \mu| < 1.001$, writing $R(s) = |\gamma(s) - \mu|$, we have

$$R'(s) < -0.79, \quad R(s) \leq R(0) = |q - \mu| < 1.001.$$

Consequently,

$$\frac{d}{ds} \log R(s) = \frac{R'(s)}{R(s)} < -\frac{0.79}{1.001} < -0.78.$$

Integration gives

$$\frac{|\gamma(s) - \mu|}{|q - \mu|} \leq e^{-0.78s}. \quad (4.27)$$

Along γ , both f' and $w - \mu$ are nonzero; hence a continuous logarithm may be chosen along the path. Integrating $f''/f' = (n-1)s_\xi$ gives

$$f'(w) = f'(q) \left(\frac{w - \mu}{q - \mu} \right)^{n-1} \exp \left((n-1) \int_q^w \left(s_\xi(v) - \frac{1}{v - \mu} \right) dv \right). \quad (4.28)$$

Let $C_0 = 10^{178}\sigma^2$. For a point at arclength s , the last exponential differs from 1 by at most

$$nC_0s e^{nC_0s}.$$

Since $C_0 < 10^{-2946}$, define the transport remainder by

$$R_q := \int_q^{w_0} \left[f'(w) - f'(q) \left(\frac{w - \mu}{q - \mu} \right)^{n-1} \right] dw.$$

Equations (4.27) and (4.28) give

$$\begin{aligned} |R_q| &\leq |f'(q)| \int_0^{\ell(\gamma)} e^{-0.78(n-1)s} nC_0s e^{nC_0s} ds \\ &\leq |f'(q)| \int_0^\infty nC_0s e^{-0.7ns} ds \\ &= \frac{C_0}{0.49n} |f'(q)| < \frac{4C_0}{n} |f'(q)|. \end{aligned}$$

Thus

$$\frac{|R_q|}{|f'(q)(q - \mu)|/n} < 10^{182}\sigma^2.$$

The exponent $0.7n$ is available because

$$-0.78(n-1) + nC_0 < -0.7n$$

for $n \geq N_*$.

The model integral itself is exact:

$$\int_q^{w_0} (w - \mu)^{n-1} dw = \frac{(w_0 - \mu)^n - (q - \mu)^n}{n}.$$

Set

$$B_q := \frac{f'(q)(w_0 - \mu)^n}{n(q - \mu)^{n-1}}, \quad C_q := f(w_0) - f(0).$$

By the arithmetic-geometric mean inequality,

$$\frac{|f'(q)|}{n} = \prod_{j=1}^{n-1} |q - \xi_j| \leq \left(\mathbb{E}|q - \xi|^2 \right)^{(n-1)/2} = \left(|q - \mu|^2 + \sigma^2 \right)^{(n-1)/2}.$$

Consequently,

$$|B_q| \leq |w_0 - \mu| \left(|w_0 - \mu| \sqrt{1 + \frac{\sigma^2}{|q - \mu|^2}} \right)^{n-1} < 0.0101(0.011)^{n-1} < (0.012)^n.$$

Similarly, for $|w| \leq 0.01$,

$$\begin{aligned} |f'(w)| &\leq n \left(\mathbb{E}|w - \xi|^2 \right)^{(n-1)/2} \\ &= n \left(|w - \mu|^2 + \sigma^2 \right)^{(n-1)/2} < n(0.011)^{n-1}. \end{aligned}$$

Integrating on the line segment from 0 to w_0 , and using $n \geq N_*$, gives

$$|C_q| \leq 0.01n(0.011)^{n-1} < (0.012)^n.$$

Since $f(q) = 0$, the definition of R_q and the exact model integral give

$$f(w_0) = B_q - \frac{f'(q)(q - \mu)}{n} + R_q.$$

As $f(w_0) = f(0) + C_q$, it follows that

$$0 = f(0) + \frac{f'(q)(q - \mu)}{n}(1 + \vartheta_q) + E_q,$$

where

$$\vartheta_q := -\frac{nR_q}{f'(q)(q - \mu)}, \quad E_q := C_q - B_q.$$

Hence

$$|\vartheta_q| \leq 10^{182}\sigma^2, \quad |E_q| < 2(0.012)^n < (0.03)^n,$$

which proves (4.24) and (4.25).

It remains to convert the endpoint identity into a potential estimate. Put $x = q - \mu$ and $u = (\xi - \mu)/x$. Then

$$|x| > 0.989, \quad |u| < 1.02, \quad |1 - u| \geq \frac{d_*}{|x|}$$

for a selected boundary zero; the separation is stronger when $q = a$. For these values,

$$|\log |1 - u| + \Re u| \leq 4 \left(\log \frac{1}{d_*} + 3 \right) |u|^2. \quad (4.29)$$

To verify this inequality, use the Taylor series when $|u| \leq 1/2$. When $|u| > 1/2$, bound its left side by $\log(1/d_*) + 3$ and use $4|u|^2 > 1$. Taking expectations and using $\mathbb{E}(\xi - \mu) = 0$ gives

$$\begin{aligned} |-U_\xi(q) - \log |q - \mu|| &\leq 4 \left(\log \frac{1}{d_*} + 3 \right) \frac{\sigma^2}{|q - \mu|^2} \\ &\leq \frac{4(\log(1/d_*) + 3)}{0.989^2} \sigma^2 < 1700\sigma^2. \end{aligned} \quad (4.30)$$

By (4.4), division of (4.24) by $f(0)$ is harmless. Set

$$A_q = \frac{f'(q)(q - \mu)}{n}, \quad \delta_q = \frac{E_q}{f(0)}.$$

Then

$$A_q(1 + \vartheta_q) = -f(0)(1 + \delta_q),$$

where

$$|\delta_q| < 16(0.03)^n \leq (0.06)^n.$$

Both ϑ_q and δ_q have modulus less than $1/2$, and therefore

$$\log |f(0)| = \log |A_q| + \log |1 + \vartheta_q| - \log |1 + \delta_q|.$$

Using $|\log |1 + u|| \leq 2|u|$ for $|u| \leq 1/2$, together with

$$\log |f'(q)| = \log n - (n-1)U_\xi(q),$$

and (4.30), gives (4.26). $\square$

The endpoint estimate immediately places the selected zeros on an effective approximate level set.

Corollary 4.6. *For every selected zero z ,*

$$|U_\xi(z) - U_\xi(a)| \leq \frac{6 \cdot 10^{182} \sigma^2 + 64(0.06)^n}{n}. \quad (4.31)$$

Proof. Apply (4.26) first to z and then to a , and use the triangle inequality. $\square$

The level-set relation will first control the mean and the distance of a from the unit circle. Put

$$Q = 10^{800}(0.06)^n.$$

The following lemma closely follows the argument of [Tao22, Proposition 5.6(i)], with the relevant estimates recorded in the quantitative form needed below.

Lemma 4.7. *One has*

$$|\mu| \leq 2 \cdot 10^5 \sigma^2 + 20Q, \quad (4.32)$$

$$d \leq 2 \cdot 10^4 \sigma^2 + 3Q. \quad (4.33)$$

Proof. For $w = a$ or for a selected zero $w = z$, put

$$G(w, \xi) = \log \left| 1 - \frac{\xi}{w} \right| + \Re \frac{\xi}{w}.$$

Here $|w| > 0.989$, $|\xi/w| < 1.02$, and

$$\left| 1 - \frac{\xi}{w} \right| = \frac{|w - \xi|}{|w|} > \frac{d_*}{|w|};$$

at $w = a$, the separation is in fact greater than $1/a$. The same two-case argument used for (4.29), now with $u = \xi/w$, gives

$$|G(w, \xi)| \leq 4 \left(\log \frac{1}{d_*} + 3 \right) \frac{|\xi|^2}{|w|^2} < 1700 |\xi|^2.$$

Since

$$U_\xi(w) = -\log |w| + \Re \frac{\mu}{w} - \mathbb{E} G(w, \xi),$$

comparison at a and z gives

$$\left| U_\xi(a) - U_\xi(z) - \left\{ \log \frac{|z|}{a} + \Re \left(\left(\frac{1}{a} - \frac{1}{z} \right) \mu \right) \right\} \right| \leq 4000V.$$

Combining this with Corollary 4.6 and using $|z| \leq 1$, we obtain

$$4000V + \frac{6 \cdot 10^{182} \sigma^2 + 64(0.06)^n}{n} \leq \left(4000 + \frac{6 \cdot 10^{182}}{n} \right) V + \frac{64 \cdot 10^{-800}}{n} Q < 10^4 V + Q.$$

Consequently,

$$\Re \left(\left(\frac{1}{a} - \frac{1}{z} \right) \mu \right) \geq \log a - E_0, \quad E_0 = 10^4 V + Q. \quad (4.34)$$

Choose angles outside H in the four intervals

$$\begin{aligned} J_1 &= [0.98\pi, 0.99\pi], & J_2 &= [1.01\pi, 1.02\pi], \\ J_3 &= [0.49\pi, 0.50\pi], & J_4 &= [1.50\pi, 1.51\pi]. \end{aligned}$$

This is possible because each interval has length 0.01π , while $|H| < 10^{-20}$. Let z_j be the corresponding zeros and set $c_j = 1/a - 1/z_j$.

For $j = 1, 2$, the imaginary parts of c_j have opposite signs. Indeed, up to an error smaller than 10^{-390} , the first two windows give

$$-1.001 < \Re \frac{1}{z_j} < -0.997,$$

while the two imaginary parts of $1/z_j$ have opposite signs and absolute values between 0.031 and 0.064. Since $1 < 1/a < 1.011$, a convex combination of c_1, c_2 is real and belongs to $[1.9, 2.1]$. Hence (4.34) and $\log(1-d) \geq -d/(1-d) \geq -1.011d$ give

$$\Re \mu \geq -\frac{1.011d + E_0}{1.9}. \quad (4.35)$$

On the other hand, the support inequality $\mathbb{E}|a - \xi|^2 > 1$ gives

$$\Re \mu \leq -d + 0.506V. \quad (4.36)$$

Comparing (4.35) and (4.36), and recalling $E_0 = 10^4V + Q$, yields

$$d \leq 1.2 \cdot 10^4V + 2Q, \quad (4.37)$$

$$|\Re \mu| \leq 1.2 \cdot 10^4V + 2Q. \quad (4.38)$$

For $j = 3, 4$, direct trigonometry gives

$$|\Re c_j| \leq 1.2, \quad \Im c_3 \geq 0.99, \quad \Im c_4 \leq -0.99.$$

For example, on these two windows

$$|\cos \theta_j| \leq \sin(0.01\pi) < 0.032, \quad |\sin \theta_j| \geq \cos(0.01\pi) > 0.999,$$

and the perturbation from $e^{i\theta_j}$ to z_j is smaller than 4κ . Applying (4.34) for both signs, followed by (4.37) and (4.38), gives

$$|\Im \mu| \leq 3.7 \cdot 10^4V + 6Q.$$

Thus

$$|\mu| \leq 6 \cdot 10^4V + 10Q. \quad (4.39)$$

Since $V = \sigma^2 + |\mu|^2 < 10^{-3124}$, the term $6 \cdot 10^4|\mu|^2$ can be absorbed into the left side of (4.39), since

$$6 \cdot 10^4|\mu| < 10^{-1557} < \frac{1}{2}.$$

It follows first that

$$|\mu| \leq 1.2 \cdot 10^5\sigma^2 + 20Q < 2 \cdot 10^5\sigma^2 + 20Q,$$

which is (4.32). Moreover,

$$1.2 \cdot 10^4|\mu|^2 \leq 1.2 \cdot 10^4 10^{-1562}(2 \cdot 10^5\sigma^2 + 20Q) < 8 \cdot 10^3\sigma^2 + Q.$$

Substitution in (4.37) now gives (4.33). $\square$

We now compare the nearby zero with the exact boundary point. The one-sided form of this comparison is essential: it retains the favorable term $-\log|z|$. The next lemma is a minor quantitative refinement of [Tao22, Proposition 5.6(ii)], obtained by making the Taylor remainder explicit.

Lemma 4.8. *For every $\theta \notin H$,*

$$U_\xi(a) - U_\xi(e^{i\theta}) \geq -10^{-20}V - Q. \quad (4.40)$$

Proof. Let z be the selected zero near $e^{i\theta}$, and define

$$G(w, \xi) = \log \left| 1 - \frac{\xi}{w} \right| + \Re \frac{\xi}{w}.$$

Direct differentiation gives the exact identity

$$|\nabla_w G(w, \xi)| = \frac{|\xi|^2}{|w|^2 |w - \xi|}. \quad (4.41)$$

The segment from z to $e^{i\theta}$ remains at distance greater than $d_*/2$ from every critical point, because its length is less than 4κ . Since

$$U_\xi(w) = -\log |w| + \Re \frac{\mu}{w} - \mathbb{E}G(w, \xi)$$

and $-\log |z| \geq 0$, the mean-value theorem and (4.41) give

$$U_\xi(z) - U_\xi(e^{i\theta}) \geq -5\kappa|\mu| - \frac{12\kappa}{d_*}V.$$

Combine this with (4.31) and (4.32). The numerical values in (4.13), together with $n \geq N_*$, make the sum of the V -terms smaller than $10^{-20}V$, while all exponential terms are bounded by Q . This proves (4.40). $\square$

The proof now closes by comparing two estimates for the second moment. Let $dm = d\theta/(2\pi)$; the set H is already π -translation-invariant. The following Fourier estimate is a quantitative version of the second-order closure argument in [Tao22, Proposition 5.6(iii) and (5.23)].

Lemma 4.9. *One has*

$$\mathbb{E}L \leq 1.01 \cdot 10^{-10}V + 1.01Q, \quad (4.42)$$

$$|\Re \mathbb{E}\xi^2| \leq 8.01 \cdot 10^{-10}V + 8Q. \quad (4.43)$$

Proof. For $0 < r < 1$, expand the potential of $r\xi$ on the unit circle and then let $r \uparrow 1$. Since the law of ξ has finite support, the logarithmic kernels converge in $L^2(dm)$. Thus all Fourier identities below hold in $L^2(dm)$, and hence almost everywhere; the finitely many possible singular angles have measure zero. In this sense,

$$U_\xi(e^{i\theta}) = \Re(\mu e^{-i\theta}) + \mathcal{R}(\theta),$$

where Parseval's identity gives

$$\|\mathcal{R}\|_2^2 = \frac{1}{2} \sum_{k=2}^{\infty} \frac{|\mathbb{E}\xi^k|^2}{k^2} \leq \frac{V^2}{2} \sum_{k=2}^{\infty} \frac{1}{k^2}.$$

Thus

$$\|\mathcal{R}\|_2 < 0.568V. \quad (4.44)$$

Jensen's formula and Fourier inversion give

$$\int U_\xi(e^{i\theta}) dm = 0, \quad (4.45)$$

$$\int \cos(2\theta) U_\xi(e^{i\theta}) dm = \frac{1}{4} \Re \mathbb{E}\xi^2. \quad (4.46)$$

Because $H + \pi = H$, the linear term $\Re(\mu e^{-i\theta})$ integrates to zero over H against each of the π -periodic weights

$$1, \quad 1 + \cos 2\theta, \quad 1 - \cos 2\theta.$$

Also, $m(H) < \Delta$. Integrating (4.40) over the complement of H , using $U_\xi(a) = -\mathbb{E}L$, (4.45), and (4.44), gives

$$(1 - m(H))\mathbb{E}L \leq 0.568\sqrt{\Delta}V + 10^{-20}V + Q.$$

This proves (4.42).

Put

$$\varepsilon := 10^{-20}V + Q, \quad W_\pm(\theta) := 1 \pm \cos(2\theta).$$

By (4.40),

$$\mathbb{E}L + U_\xi(e^{i\theta}) \leq \varepsilon \quad (\theta \notin H).$$

Since $\mathbb{E}L \geq 0$ and $W_\pm \geq 0$, integration over H^c gives

$$\int_{H^c} W_\pm(\theta) U_\xi(e^{i\theta}) dm \leq \varepsilon \int_{H^c} W_\pm(\theta) dm.$$

By (4.45) and (4.46),

$$\int W_\pm(\theta) U_\xi(e^{i\theta}) dm = \pm \frac{1}{4} \Re \mathbb{E} \xi^2.$$

Because $H + \pi = H$, the linear term $\Re(\mu e^{-i\theta})$ integrates to zero over H against $W_\pm$. Therefore

$$\pm \frac{1}{4} \Re \mathbb{E} \xi^2 \leq \varepsilon + \left| \int_H W_\pm(\theta) \mathcal{R}(\theta) dm \right|.$$

Furthermore,

$$\|W_\pm \mathbf{1}_H\|_2 \leq 2\sqrt{m(H)} < 2\sqrt{\Delta}.$$

Thus, by (4.44),

$$\frac{1}{4} |\Re \mathbb{E} \xi^2| \leq 2(0.568)\sqrt{\Delta}V + \varepsilon.$$

The numerical values of Δ and ε now imply

$$|\Re \mathbb{E} \xi^2| \leq 8.01 \cdot 10^{-10}V + 8Q,$$

which is (4.43). $\square$

The geometry of $\mathcal{A}$ supplies the incompatible estimate. The next lemma follows Tao's arc-projection and negative second-moment argument in [Tao22, pp. 389–390], with a minor adjustment to the absorption step based on Young's inequality and with all constants made explicit.

Lemma 4.10. *One has*

$$\Re \mathbb{E} \xi^2 \leq \left(-\frac{1}{4} + 4.04 \cdot 10^{-8} \right) V + 576d^2 + 404Q. \quad (4.47)$$

Proof. Choose $y \in \mathcal{A}$ as in Lemma 4.2. If $y = 0$, then

$$\Re y^2 = 0 = -\frac{1}{2}|y|^2.$$

Otherwise write $y = re^{i\phi}$. Since $y \in \mathcal{A}$, we have $r = 2\cos\phi$, and $0 < r \leq 1$ implies

$$|\phi| \in [\pi/3, \pi/2].$$

Consequently, in all cases,

$$\Re y^2 \leq -\frac{1}{2}|y|^2.$$

Write $\xi = y + e$. Using $|\xi| \leq |y| + |e|$, we obtain

$$\begin{aligned} \Re \xi^2 + \frac{1}{4}|\xi|^2 &\leq -\frac{1}{2}|y|^2 + 2|y||e| + |e|^2 + \frac{1}{4}(|y| + |e|)^2 \\ &= -\frac{1}{4}(|y| - 5|e|)^2 + \frac{15}{2}|e|^2 \\ &\leq \frac{15}{2}|e|^2. \end{aligned}$$

Since $|e| \leq 6(d + L)$, it follows that

$$\Re \xi^2 \leq -\frac{1}{4}|\xi|^2 + 270(d + L)^2 \leq -\frac{1}{4}|\xi|^2 + 576(d^2 + L^2).$$

Now $L^2 \leq (\log 2)L$. Taking expectations and applying (4.42) gives (4.47). $\square$

Combining the preceding quantitative estimates yields the following effective version of [Tao22, Theorem 5.1].

Proposition 4.11. *No polynomial satisfying (4.1) and (4.2) exists when $n \geq N_*$.*

Proof. Suppose first that

$$\sigma^2 \leq 10^{1000}(0.06)^n.$$

(4.33) gives

$$d < 3 \cdot 10^{1004}(0.06)^n < 8^{-n},$$

contrary to (4.2).

It remains to consider $\sigma^2 > 10^{1000}(0.06)^n$. Then

$$\eta := \frac{Q}{\sigma^2} < 10^{-200}.$$

By (4.32),

$$|\mu| \leq (2 \cdot 10^5 + 20\eta)\sigma^2 < 2.01 \cdot 10^5 \sigma^2.$$

Since $\sigma^2 \leq V$ and (4.6) gives $V < 10^{-3124}$, it follows that

$$\frac{V}{\sigma^2} = 1 + \frac{|\mu|^2}{\sigma^2} \leq 1 + (2.01 \cdot 10^5)^2 \sigma^2 < 1.001.$$

Likewise, (4.33) gives

$$d \leq (2 \cdot 10^4 + 3\eta)\sigma^2 < 2.01 \cdot 10^4 \sigma^2.$$

Hence, using $V \geq \sigma^2$ and again (4.6),

$$\frac{576d^2}{V} \leq 576(2.01 \cdot 10^4)^2 \sigma^2 < 10^{-1400}.$$

The geometric estimate (4.47) therefore implies

$$\Re \xi^2 < -0.2499999V.$$

The Fourier estimate (4.43), on the other hand, implies

$$\Re \xi^2 > -8.02 \cdot 10^{-10}V.$$

These inequalities are incompatible. $\square$

The compactness argument near the unit circle has therefore been replaced by finite obstacle removal and explicit estimates. We now combine the two endpoint theorems with the quantitative results already in the literature.

5. ASSEMBLY OF THE RANGES

We first record that the explicit constants used at the two endpoints are compatible:

$$N_{\text{org}} < 10^{368} < N_*. \quad (5.1)$$

Indeed, the base in (3.3) is less than 10^{23} . The following proof now exhausts all possible positions of the distinguished zero.

Proof of Theorem 1.2. Suppose to the contrary that $n \geq N_*$ and that f is a counterexample. Multiplication by a scalar and rotation preserve the problem, so f may be taken monic and its distinguished zero may be taken to be $a \in [0, 1]$. Assume that f' has no zero in $\overline{D(a, 1)}$, and put $d = 1 - a$.

If $a = 0$, the Gauss–Lucas theorem places every critical point in $\overline{\mathbb{D}} = \overline{D(a, 1)}$, a contradiction. If $0 < a \leq n^{-1/8}$, then (5.1) and Theorem 3.1 give the same contradiction.

It remains to suppose that $a > n^{-1/8}$. If $d \geq n^{-1/64}$, then

$$na^7 d^4 \geq n^{1-7/8-4/64} = n^{1/16} > 20800.$$

Equivalently,

$$n > \frac{20800}{a^7(1-a)^4},$$

so Chalebgwa's theorem [Cha20, Theorem 1.1] applies and rules out this case.

We are left with $d < n^{-1/64}$. If $d \geq 8^{-n}$, Proposition 4.11 rules out the counterexample. Finally, suppose that $0 \leq d < 8^{-n}$. For every $n \geq 64$,

$$2n^9 \leq 2^n,$$

because it holds at $n = 64$ and the ratio $2^n/(2n^9)$ is increasing thereafter. Hence

$$8^{-n} \leq \frac{1}{2n^9 4^n}.$$

Thus

$$a > 1 - \frac{1}{2n^9 4^n},$$

and Chijiwa's explicit boundary theorem [Chi11, Main theorem] applies. More precisely, it supplies a zero ζ of f' for which

$$|\zeta - a| \leq 1 - c_n(1 - a),$$

where Chijiwa's explicit congruence-dependent coefficient c_n is nonnegative for $n \geq 9$. Hence $|\zeta - a| \leq 1$. Every possible value of a has led to a contradiction. $\square$

Remark 5.1. The theorem identifies a valid universal threshold, not the smallest one. The numerical value is dominated by the decision to separate all finite obstacles using fixed scales and by the high Fourier cutoff $M = 10^{410}$. None of the inequalities in the proof is close to equality.

APPENDIX A. NUMERICAL AUDIT

For completeness, Table A.1 gathers the numerical implications of $n \geq 10^{200000}$ that were used in the proof. Each bound in the table follows by taking base-ten logarithms; the estimates are intentionally weakened to leave margins of several orders of magnitude.

TABLE A.1. Numerical margins in the effective proof

Quantity	Bound used	Consequence
$n^{-1/64}$	$\leq 10^{-3125}$	$5n^{-1/64} < 10^{-3124}$
$n^{1/16}$	$\geq 10^{12500}$	$n^{1/16} > 20800$
N_{org}	$< 10^{368}$	$N_{\text{org}} < N_*$
$D(\nu)$	$< 10(10^{-410} + 10^{-1562} + 10^{-3119} + 10^{-199589})$	$< \kappa/(100\pi)$
$ H $	$< 6 \cdot 10^{-21}$	$< \Delta$
κ/d_*	$= 5 \cdot 10^{-229}$	$< 10^{-20}$
Q/σ^2 in the second case	$< 10^{-200}$	$V < 1.001\sigma^2$

We spell out three checks that involve sums rather than a single monomial comparison. First, the exceptional-angle construction gives

$$|H| < 2 \left(20 \cdot 10^{-22} + 40 \cdot 10^{32} \cdot 10^{-55} + \frac{32 \cdot 10^{32} \cdot 2.5 \cdot 10^{-172}}{10^{-30}} + 2000(10^{32} + 1)10^{-400} \right) < 6 \cdot 10^{-21}.$$

Second, in Lemma 4.5, the two absolute errors arising at the inner endpoint are each bounded by 0.012^n ; hence

$$2(0.012)^n < 0.03^n.$$

Third, in the small-variance alternative of Proposition 4.11,

$$\frac{3 \cdot 10^{1004}(0.06)^n}{8^{-n}} = 3 \cdot 10^{1004}(0.48)^n < 1.$$

These checks also show why a decimal used as an upper bound must not be rounded downward; every displayed decimal used as an upper bound has been rounded upward.

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